

Udiptaman Das

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PhD Candidate in Computer Science with expertise in multimodal machine learning, large language models, and structured knowledge engineering. Experienced in architecting robust, evaluation-first AI systems for complex reasoning tasks, with demonstrated success deploying end-to-end solutions across high-stakes domains.

TECHNICAL SKILLS

Languages & Software Engineering: Python, SQL, C++, Java, R, Bash; Git, Docker, REST APIs, unit testing, code review; PostgreSQL, MongoDB.

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers; LLM systems — LangChain, CrewAI, AutoGen, RAG, LoRA/QLoRA fine-tuning; computer vision (segmentation, 3D/4D medical imaging); statistical modeling, uncertainty quantification, evaluation-first development.

Data & Cloud: Knowledge representation & graph data (Neo4j, SPARQL, RDF; ontology alignment incl. SNOMED CT, ICD-10, LOINC); pandas, NumPy; visualization (Plotly, matplotlib, seaborn, Streamlit); AWS (S3, Lambda, SageMaker), GCP.

EDUCATION

University of Missouri, Kansas City PhD Candidate, Computer Science Passed Comprehensive Examination — May 28, 2026	Kansas City, USA Expected May 2027
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University of Missouri, Kansas City <i>MS in Data Science and Analytics</i>	Kansas City, USA July 2024
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Indian Institute of Technology <i>Foundational level degree in Programming and Data Science</i>	Madras, India Dec 2021
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Assam Engineering College <i>BE in Electrical Engineering</i>	Assam, India Feb 2020
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RESEARCH EXPERIENCE

University of Missouri, Kansas City
Doctoral Researcher (Advisor: Dr. Yugyung Lee) Aug 2023 – Present

Trustworthy multimodal knowledge graphs (dissertation core): The core of my dissertation: collaborating LLM agents transform unstructured clinical text, imaging, and segmentation into ontology-aligned multimodal knowledge graphs; vision KGs ground image-derived evidence in structured clinical context; and retrieval-grounded, uncertainty-calibrated validation (TrustKG) makes the graphs reliable enough for explainable reasoning and decision support.

KG-grounded oncology reasoning (abdominal CT): Applying the framework to GI cancers on abdominal CT: ontology-grounded imaging-phenotype KGs enabling explainable oncology reasoning and surgical decision support — a multi-institutional collaboration with NKC Health and the SWOG Radiation Oncology Committee (ongoing).

Airway digital-twin MMKG (Project GENESIS, ongoing): Extending the framework to 4D (3D + temporal) segmentation data: airway-anatomy knowledge graphs for digital-twin modeling and clinical decision support, in collaboration with Children’s Mercy Hospital.

Large-scale corporate disclosure mining engine: Designed the ML stack for a financial-research collaboration with CUNY Brooklyn, extracting mission/strategy, talent-risk, and stock-compensation disclosures from SEC 10-K filings. Built a FIBO-guided, retrieval-first pipeline — hybrid retrieval (BM25 + dense embeddings with reciprocal rank fusion and cross-encoder reranking), NLI validation, and a LoRA-adapted quantized LLM for field extraction, with anomaly handling for auditability.

Community health web platform (Our Health Jackson County): Contributed to the development of [WeListen](#), a React-based web platform supporting regional health-disparity initiatives through community feedback and data collection.

WORK EXPERIENCE

Graduate Teaching Assistant, University of Missouri–Kansas City (Aug 2024 – May 2026) — C++ and Java course instruction, lab support, and homework grading.

Student Assistant, University of Missouri–Kansas City (Jan 2023 – Dec 2023) — Operational and inventory support across campus labs.

Data Science Intern, Deepneura Technologies (Mar 2022 – May 2022) — Object detection in gaming video and signature detection in document images.

Junior Data Engineer, Dibwas Websoft Technologies (May 2021 – Oct 2021) — Built and tuned ML models; SHAP-based KPI analysis and Python-based data visualization.

PUBLICATIONS

1. **Das, U.**, et al. “Failure-Aware Tool Orchestration for Verifiable Medical Imaging Agents.” Workshop on AI Agents for Biomedical Imaging and Multimodal Clinical Data (AIM), NeurIPS 2026 — Oral.
2. **Das, U.**, et al. “Reasoning Across Modalities: An Evidence-Aware Multimodal Clinical Agent for CT Imaging and Longitudinal Clinical Records.” Workshop on AI Agents for Biomedical Imaging and Multimodal Clinical Data (AIM), NeurIPS 2026.
3. **Das, U.**, et al. “Ontology-Grounded Imaging Phenotype Knowledge Graphs for Explainable Oncology Reasoning from Abdominal CT.” Accepted at AMIA 2026 Annual Symposium.
4. Atmakuri, K. B., **Das, U.**, et al. “NS-CARE: Risk-Aware Neuro-Symbolic Clinical Decision Support for Safe Medication Recommendation from Electronic Health Records.” Accepted at AMIA 2026 Annual Symposium.
5. **Das, U.**, et al. “From Perception to Reasoning: Image-Grounded Knowledge Graphs for Topology-Aware Medical Analysis in Abdominal CT.” 1st Workshop on Medical Reasoning (MedReasoner), CVPR 2026.
6. **Das, U.**, et al. “Clinical Knowledge Graph Construction and Evaluation with Multi-LLMs via Retrieval-Augmented Generation.” arXiv:2601.01844.
7. Ho, D. H., **Das, U.**, et al. “Leveraging Multi-Agent Systems and Large Language Models for Diabetes Knowledge Graphs.” IEEE Big Data Conference — 4th Workshop on Knowledge Graphs and Big Data.

Under Review

8. **Das, U.**, et al. “Airway-MMKG: Knowledge-Guided Multimodal Reasoning for Airway Structural Recovery and Abnormality Retrieval.” Under review, WACV 2027 (Applications Track).
9. **Das, U.**, et al. “TRUST-KG: Evidence-Grounded, Calibrated, and Selective Knowledge Graph Construction from Heterogeneous Clinical Big Data.” Under review, IEEE Big Data 2026.